Data Management
02 Data modeling

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Computer Science and Biomedical Engineering
Institute of Human-Centred Computing

Slides credit: Dominik Kowald - Matthias Boehm

Course calendar

- Part 1: Data Modeling and relational databases
  - (1) Mon. 02.03.2026: Overview of databases
  - (2) Mon. 09.03.2026: Data modeling + **Exercise 1 presentation**
  - (3) Mon. 16.03.2026: Relational databases and normalization (+ bit of SQL for Ex. 1)
  - **No lecture on Mon. 23.03.2026 (time for Ex1 + VO exam #1)**
  - (4) Mon. 30.03.2026: Query languages (SQL) + **Exercise 2 presentation**
  - **Easter Break**
  - Mon. 20.04.2026: no lecture (time for Ex2)
  - (5) Mon. 27.04.2026: Relational algebra and query processing

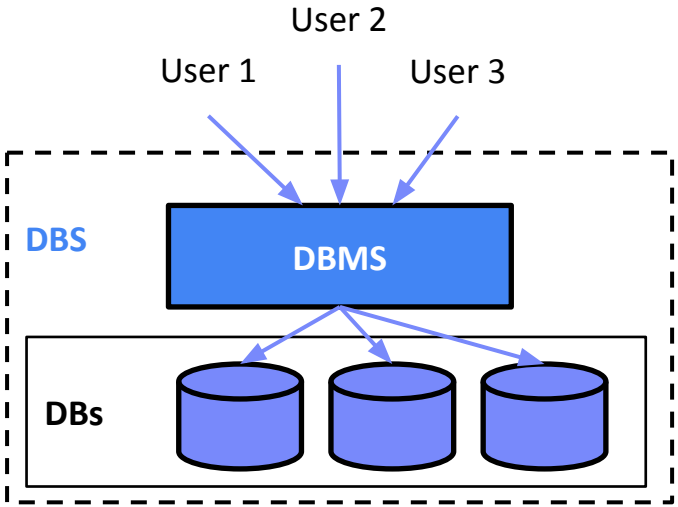
Last week

- Introduction to databases
- Course organization

Last week

- **Introduction to databases**
- **Course organization**

- **What is a database system?**
  - Database system (DBS): DBMS + DBs
  - DBMS: Database Management System (Software to handle DBs)
  - DBs: Database (data/metadata collection to describe something)
  - Note: DB also a short for DBS/DBMS



This week: Agenda

- Database design
- Entity-relationship (ER) models and diagrams

This week: Agenda

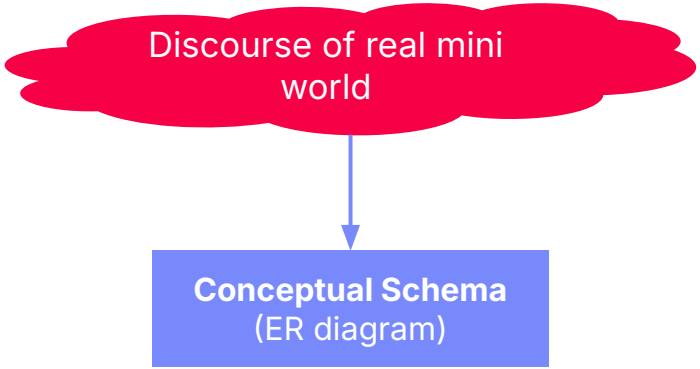
- Database design
- Entity-relationship (ER) models and diagrams

- Exercise 1: data modeling

Database design

Data Modeling

- Concept for describing data objects and their relationships
- Schema: Description (structure, semantics) of specific data collection

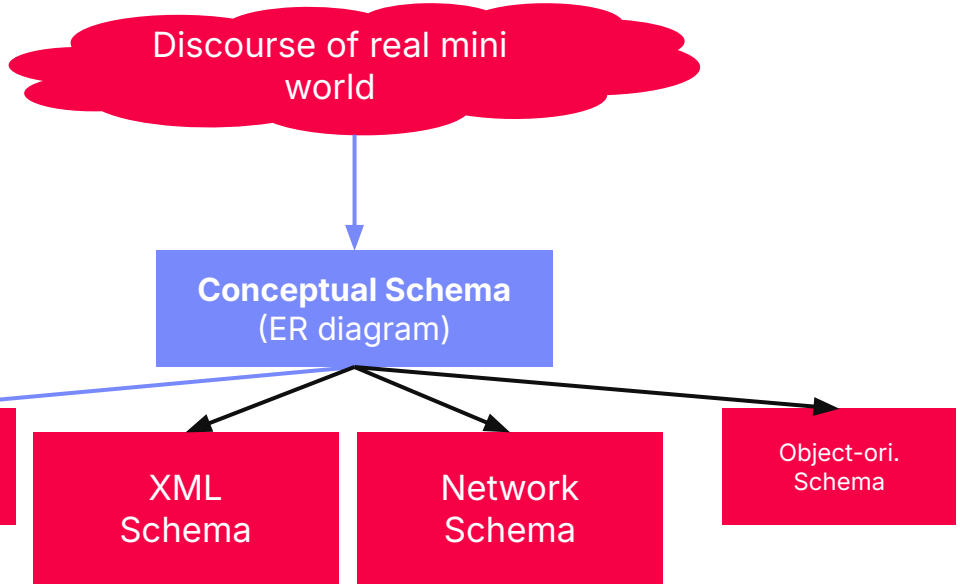


This lecture

Manual Modeling

Data Modeling

- Concept for describing data objects and their relationships
- Schema: Description (structure, semantics) of specific data collection



This lecture

Next lecture

Manual Modeling

Semi-automatic Transformation

Data Models

- **Conceptual Data Models**
  - **Entity-Relationship Model (ERM)**, focus on data, ~1975
  - Unified Modeling Language (UML), focus on data and behavior (e.g., object-oriented class diagram), ~1990

Data Models

- **Conceptual Data Models**
  - **Entity-Relationship Model (ERM)**, focus on data, ~1975
  - Unified Modeling Language (UML), focus on data and behavior (e.g., object-oriented class diagram), ~1990

- **Logical Data Models**
  - **Relational data model**
  - Key-value
  - Document (XML, JSON)
  - Graph
  - Time series
  - …

DB Design Lifecycle Phases

#1 Requirements engineering

- Collect and analyze data and application requirements
- **Specification documents**

DB Design Lifecycle Phases

#1 Requirements engineering

- Collect and analyze data and application requirements
- Specification documents

#2 Conceptual Design (this lecture, exercise 1)

- Model data semantics and structure, independent of logical data model
- ER model / diagram

DB Design Lifecycle Phases

#1 Requirements engineering

- Collect and analyze data and application requirements
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#2 Conceptual Design (this lecture, exercise 1 and project)

- Model data semantics and structure, independent of logical data model
- ER model / diagram

#3 Logical Design (next lecture, exercise 1)

- **Model data with implementation details of concrete data model**
- **e.g., relational schema + integrity constraints, permissions, etc**

DB Design Lifecycle Phases

#1 Requirements engineering

- Collect and analyze data and application requirements
- Specification documents

#2 Conceptual Design (this lecture, exercise 1 and project)

- Model data semantics and structure, independent of logical data model
- ER model / diagram

#3 Logical Design (next lecture, exercise 1 and project)

- Model data with implementation details of concrete data model
- e.g., relational schema + integrity constraints, permissions, etc

#4 Physical Design

- **Model user-level data organization in a specific DBMS (and data model)**
- Account for **deployment environment** and **performance requirements**

Tool Support

- #1 **Visual Design Tools**
  - Draw ER diagrams in any presentation software
    (e.g., MS PowerPoint, LibreOffice)
  - Many desktop or web-based tools support ER diagrams directly
    (e.g., MS Visio, creately.com)

- #2 **Design Tools w/ Code Generation**
  - Draw and validate ER diagrams
  - Generate relational schemas as SQL data definition language scripts
  - **Examples:** SAP (Sybase) PowerDesigner, MS Visual Studio plugins (SQL server), etc.

Note: For the exercise, please use the basic drawing tool draw.io

Entity-relationship (ER) models and diagrams

Chen Notation



[Peter P. Chen: The Entity-Relationship Model - Toward a Unified View of Data. **ACM Trans. Database Syst. 1(1) 1976**]

[Peter P. Chen: The Entity-Relationship Model: Toward a Unified View of Data. **VLDB 1975**]



ER Diagram Components (Chen Notation)

- **Entity (noun)**
  - Entities are objects of the real world
  - An entity instance is a concrete instantiation of an entity
  - Rectangular shape + capitalized



Example: Entity Instance

Mr	**Iacono**, Lucas Emanuel, PhD
Email	liacono@tugraz.at
Homepage	https://lucasiacono.com/
Telephone TU Graz	-
Mobile phone	4331687332824
Office hours	09:00 - 17:00
Additional info	In case you require consultation hours please coordinate in advance via e-mail
Postal address	7060 Institute of Human-Centred Computing 8010 Graz, Sandgasse 36/III



ER Diagram Components (Chen Notation)

- **Entity (noun)**
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- **Relationship (verb)**
  - Relationships are concrete associations of entities
  - Diamond shape + **non-c**apitalized



ER Diagram Components (Chen Notation)

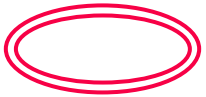
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  - Rectangular shape + capitalized
- **Relationship (verb)**
  - Relationships are concrete associations of entities
  - Diamond shape + non-capitalized
- **Attribute**
  - Entities or relationships are characterized by attribute-value pairs
  - **Round shape + capitalized**
  - Multi-valued attributes (e.g., phone numbers)







Multi-valued attributes



ER Diagram Components (Chen Notation)

- **Keys**
  - Attributes that **uniquely identify** an entity
  - **Every entity** type **must have** such **a key (mandatory!)**
  - Natural or surrogate (artificial) keys

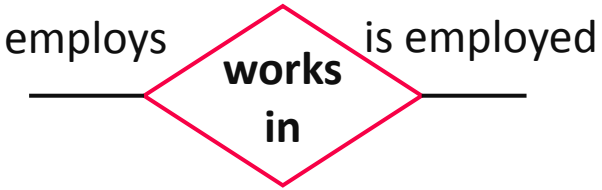


ER Diagram Components (Chen Notation)

- **Keys**
  - Attributes that uniquely identify an entity
  - Every entity type must have such a key **(mandatory!)**
  - Natural or surrogate (artificial) keys



- **Role**
  - **Optional** description of relationship types
  - Useful for recursive relationships
  - E.g., an employee is employed in a project but a project employs an employee



An Employee DB Example

[Peter P. Chen: The Entity-Relationship Model - Toward a Unified View of Data. **ACM Trans. Database Syst. 1(1) 1976**]



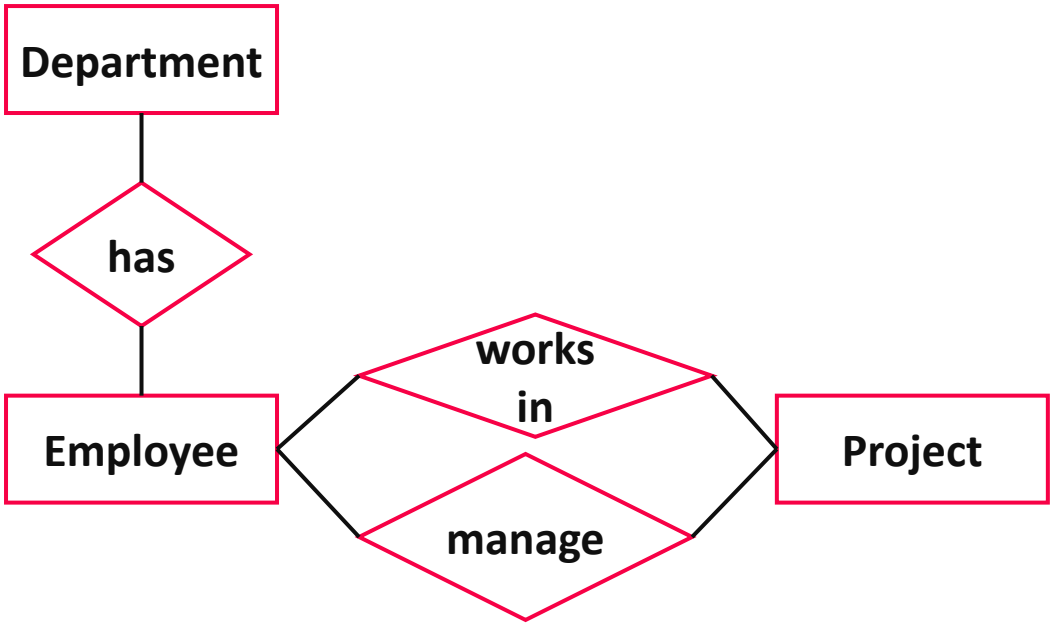
Department

Employee

Project

An EmployeeDB Example

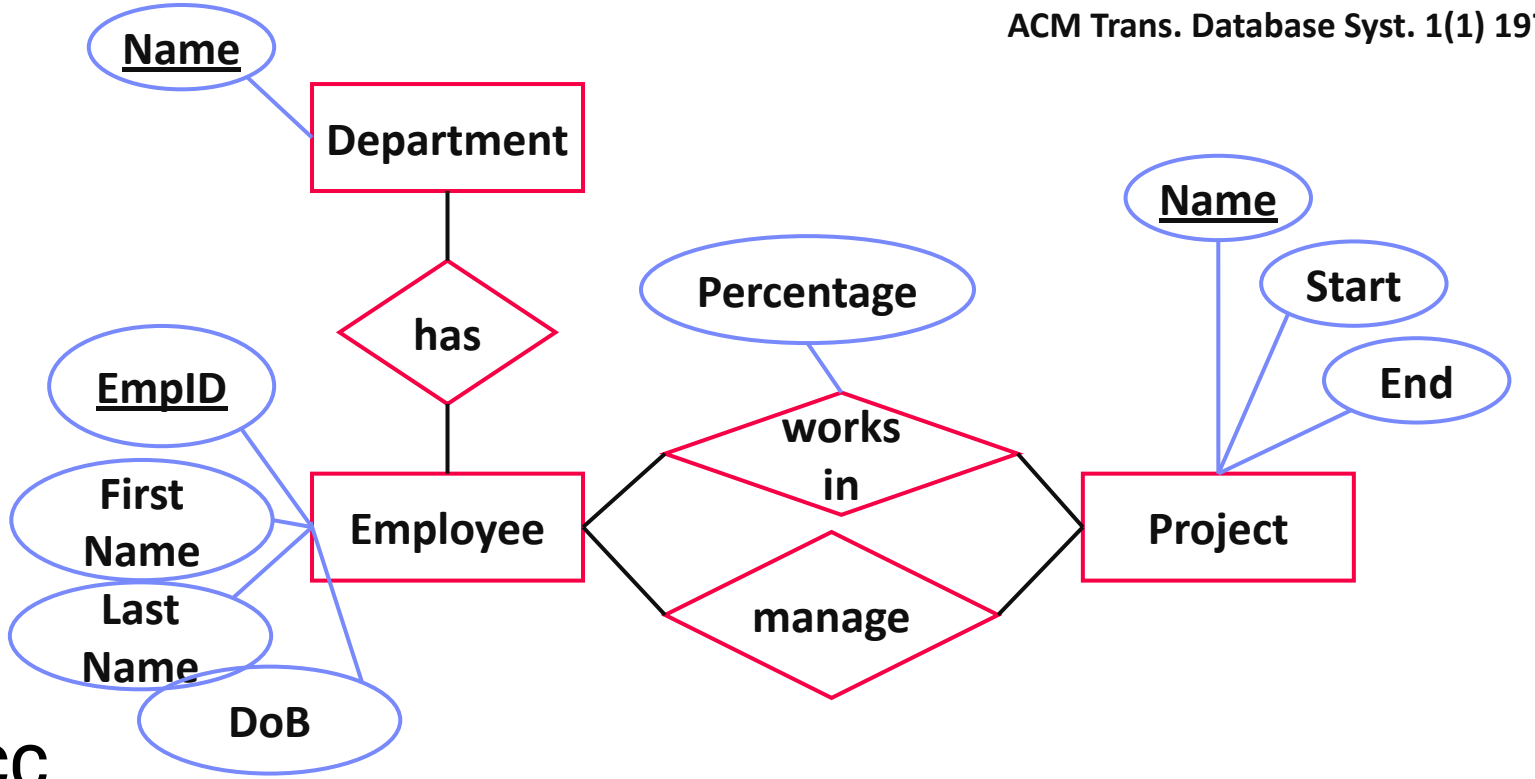
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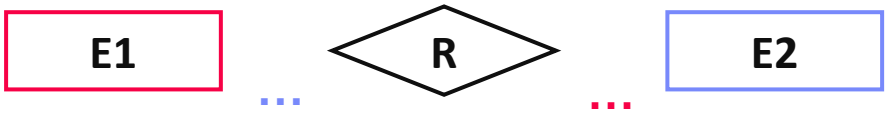
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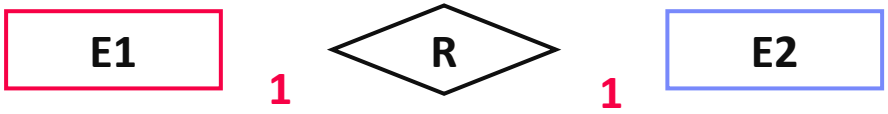




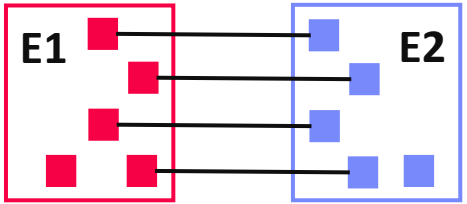
Multiplicity/Cardinality in Chen Notation



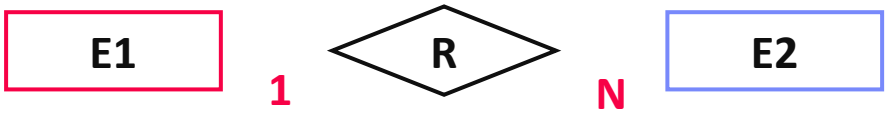
Multiplicity/Cardinality in Chen Notation



- 1:1 (one-to-one)
  - Each e1 relates to at most one e2
  - Each e2 relates to at most one e1
  - E.g., 1 employee can manage 1 department and 1 department can be managed by 1 employee

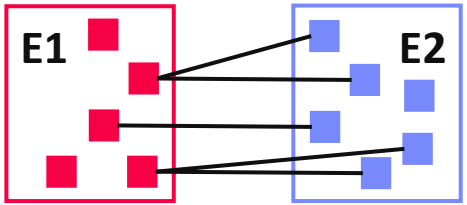


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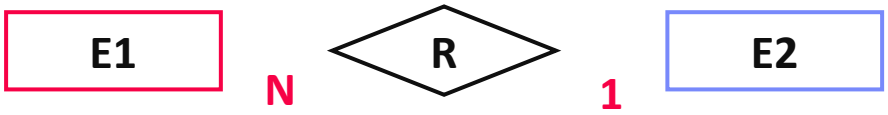


- 1:N (one-to-many)
  - Each e1 relates to many e2 (0,1,…N)
  - Each e2 relates to at most one e1
  - E.g., 1 **department** can have N **employees**

  but 1 **employee** can only have **1 department**

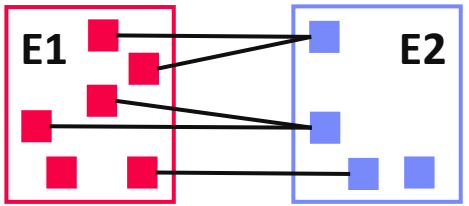


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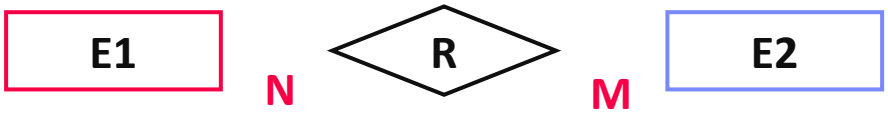


- N:1 (many-to-one)
  - Each e2 relates to many e1 (0,1,…N)
  - Each e1 relates to **at most** one e2
  - E.g., 1 employee can manage N projects

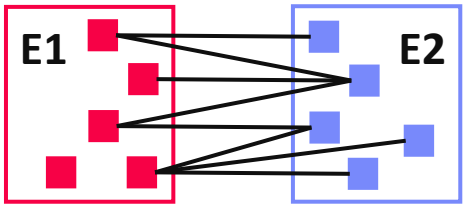
  but 1 project can be managed by just 1 employee



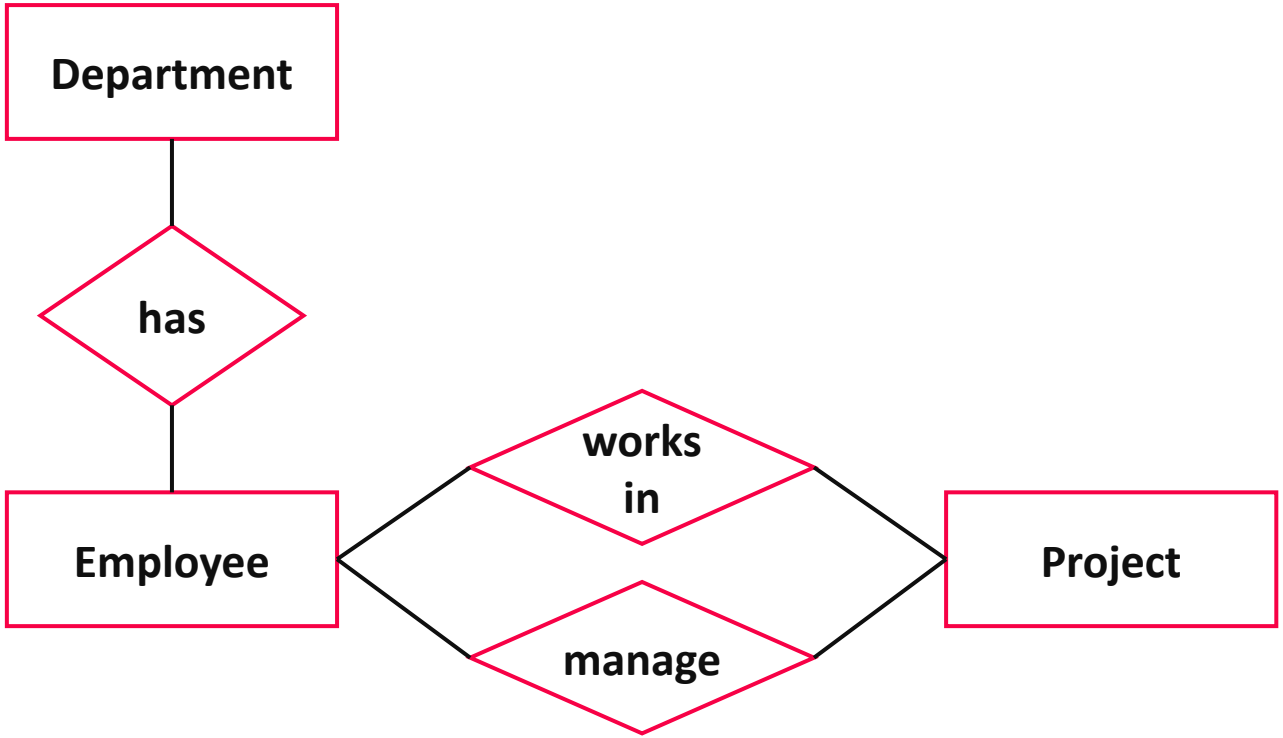
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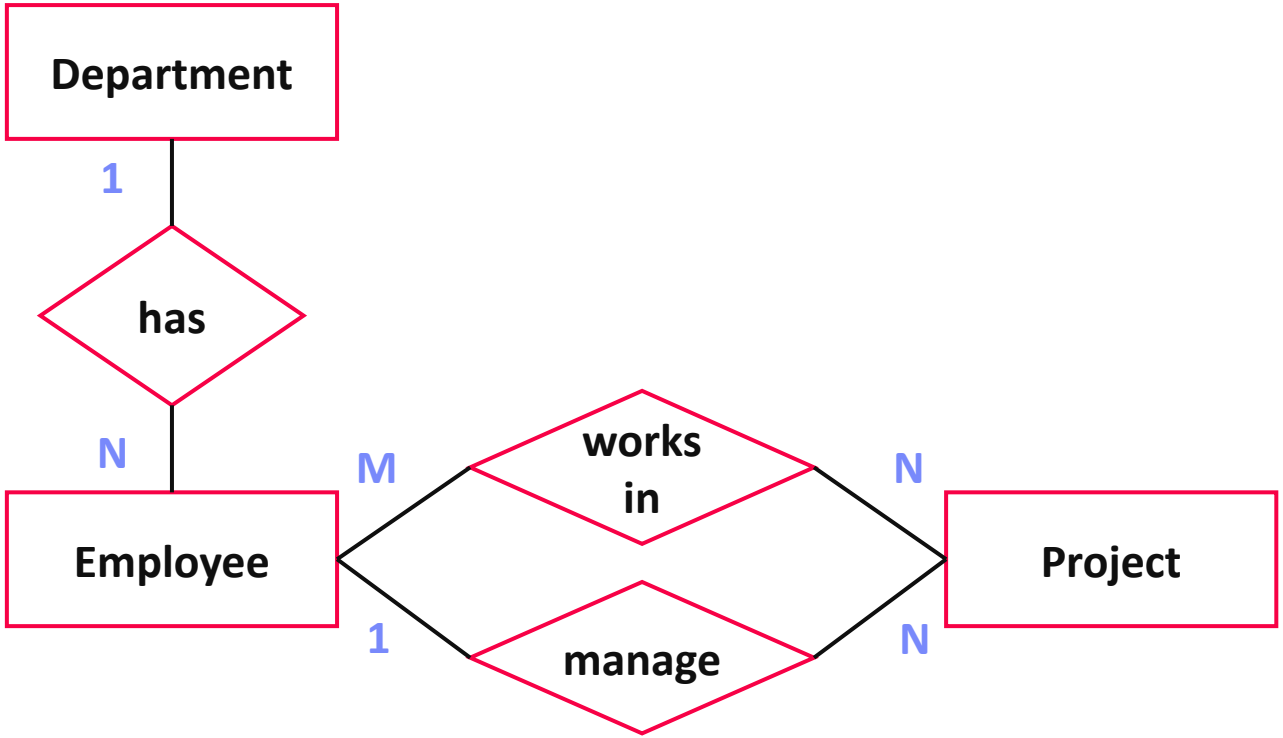
- N:M (many-to-many)
  - Each e1 relates to many e2 (0,1,…M)
  - Each e2 related to many e1 (0,1,…N)
  - e.g., 1 employee can work in N projects and 1 project can have M employees



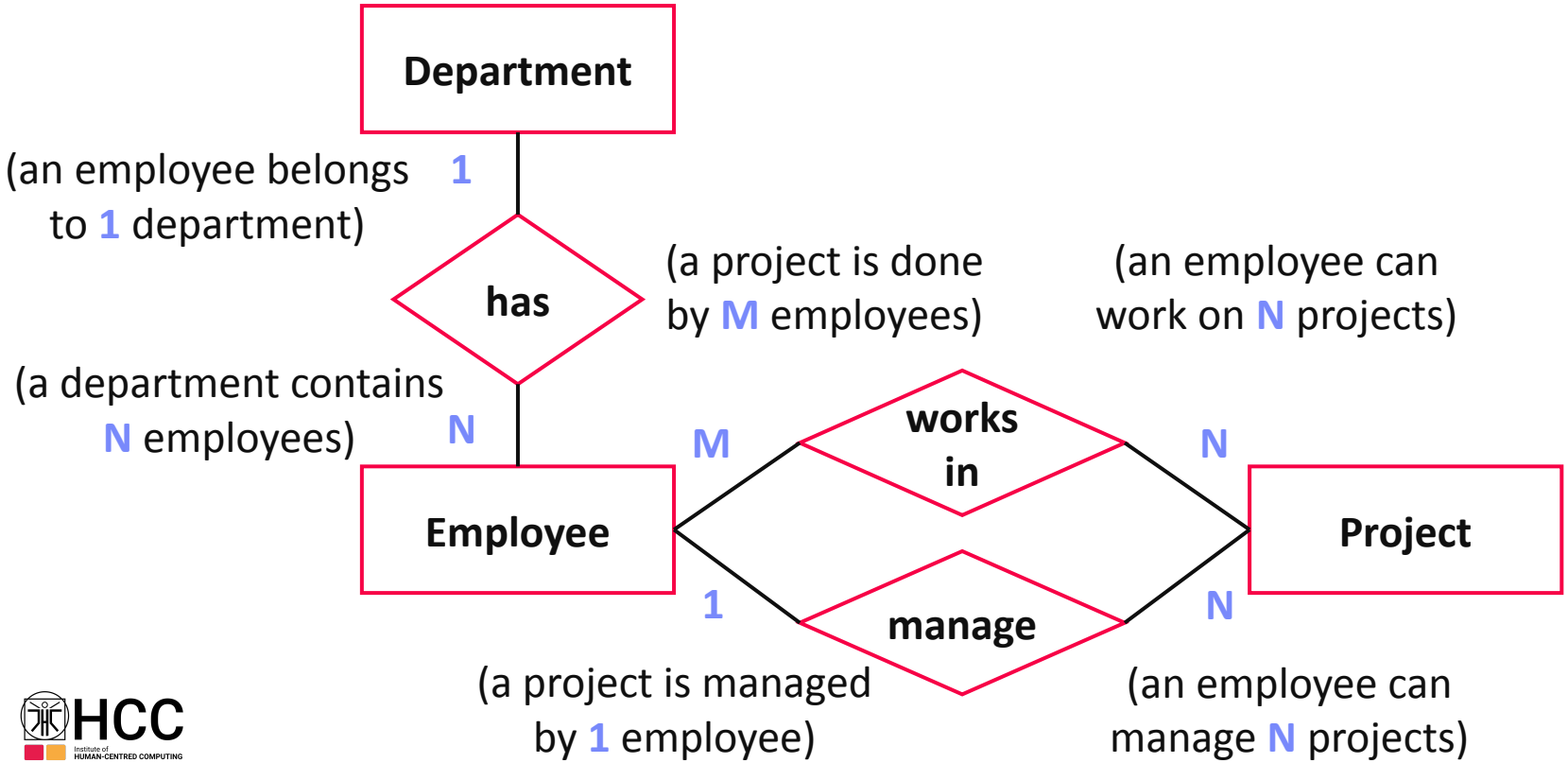
An EmployeeDB Example



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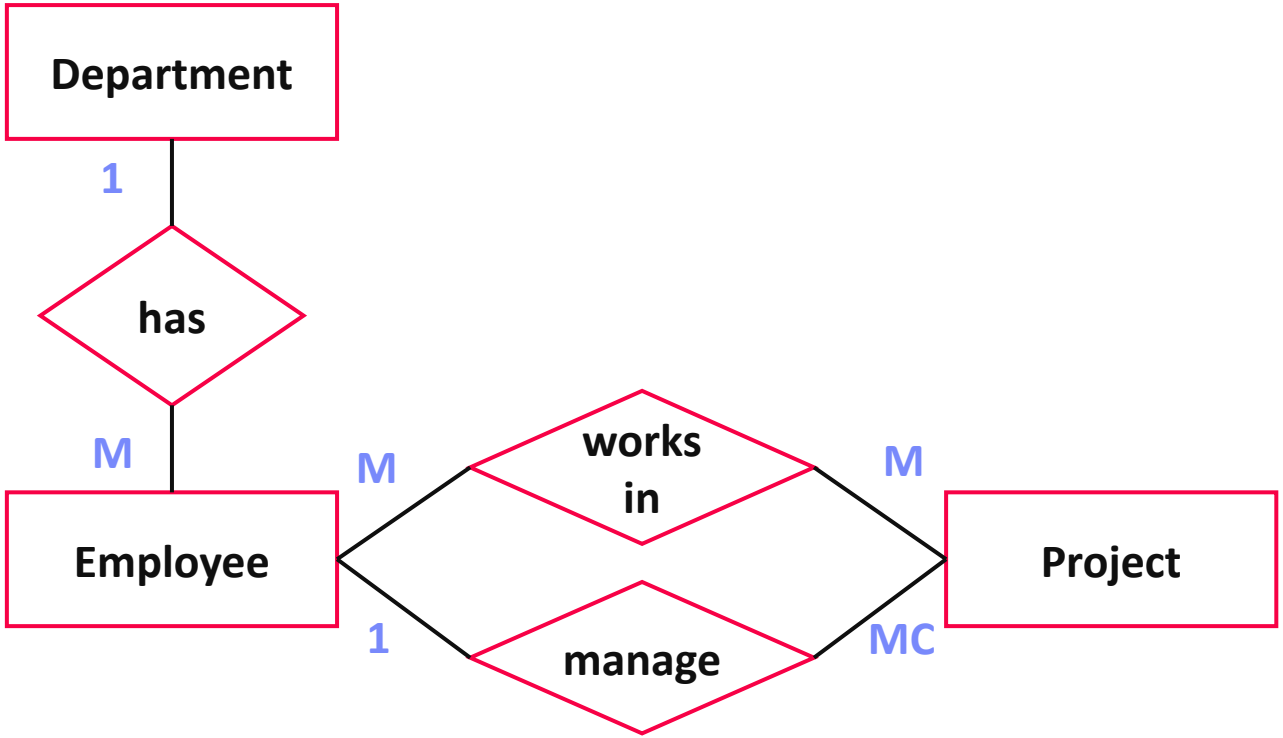
Multiplicity/Cardinality in Modified Chen Notation

- **Extension:** C (“choice”/“can”) to model 0 or 1, while 1 means exactly 1 and **M means at least 1** (MC could also be 0)

Multiplicity/Cardinality in Modified Chen Notation

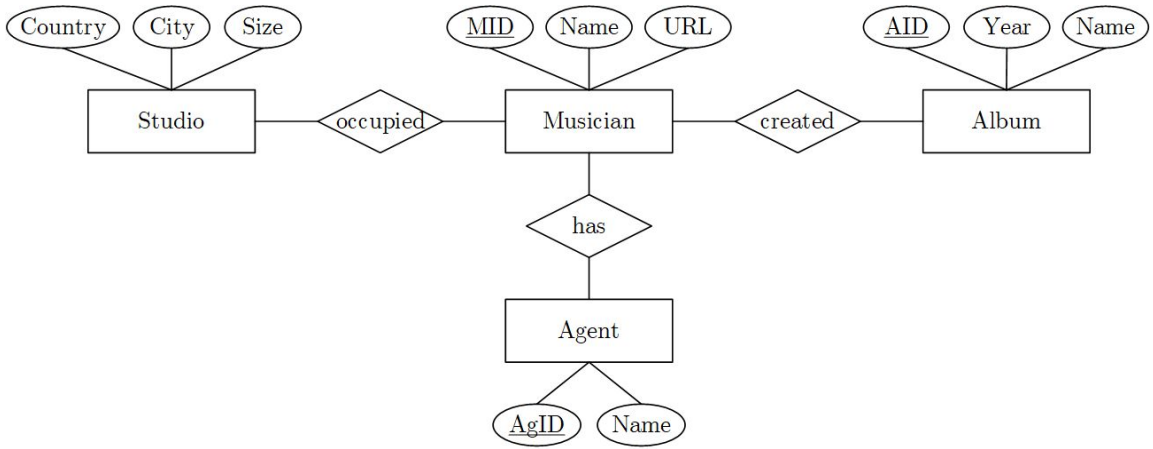
- Extension: C (“choice”/“can”) to model 0 or 1, while 1 means exactly 1 and M means at least 1 (MC could also be 0)
  - 1:1 – [1] to [1]
  - 1:C – [1] to [0 or 1]
  - 1:M – [1] to [at least 1]
  - 1:MC – [1] to [arbitrary many] (also 0 is possible!)
  - C:C – [0 or 1] to [0 or 1] 1:1 in Chen
  - C:M – [0 or 1] to [at least 1]
  - C:MC – [0 or 1] to [arbitrary many] 1:N in Chen
  - MC:C – [arbitrary many] to [0 or 1] N:1 in Chen
  - M:M – [at least 1] to [at least 1]
  - M:MC – [at least 1] to [arbitrary many]
  - MC:MC – [arbitrary many] to [arbitrary many] M:N in Chen (Allows everything)

An EmployeeDB Example in Modified Chen Not.

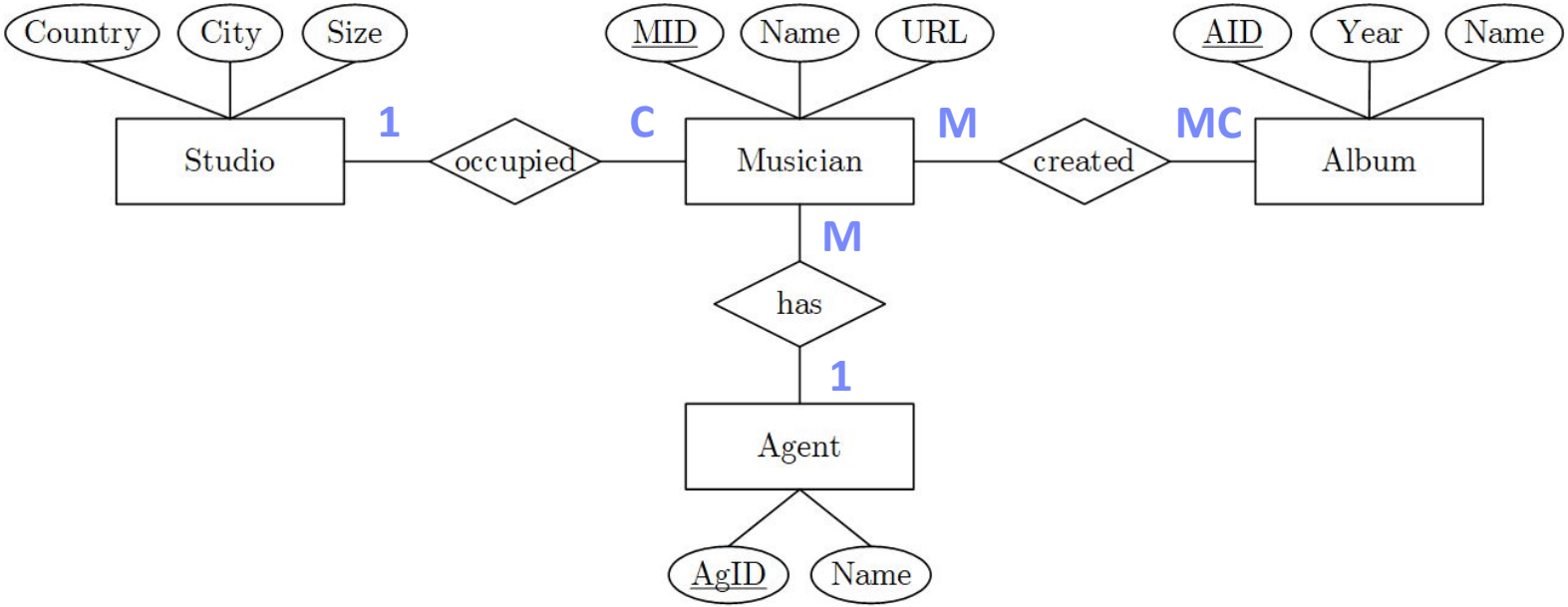


Another Example

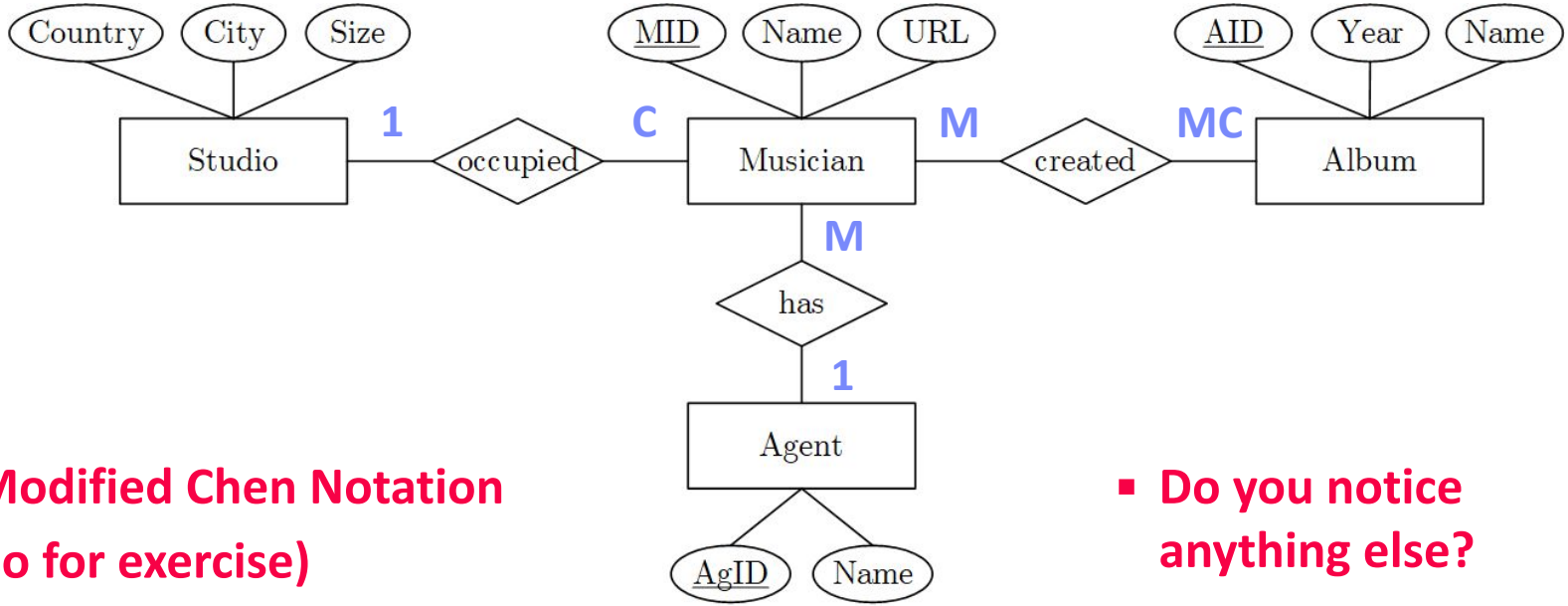
- Task: Cardinalities in **Modified-Chen Notation**
  - A musician might have created none or arbitrary many albums, and any album is created by at least one musician.
  - Every musician has exactly one agent, and an agent might be responsible for one to ten musicians.
  - Every musician occupies exactly one studio, and musicians never share a studio.



Another Example: Solution

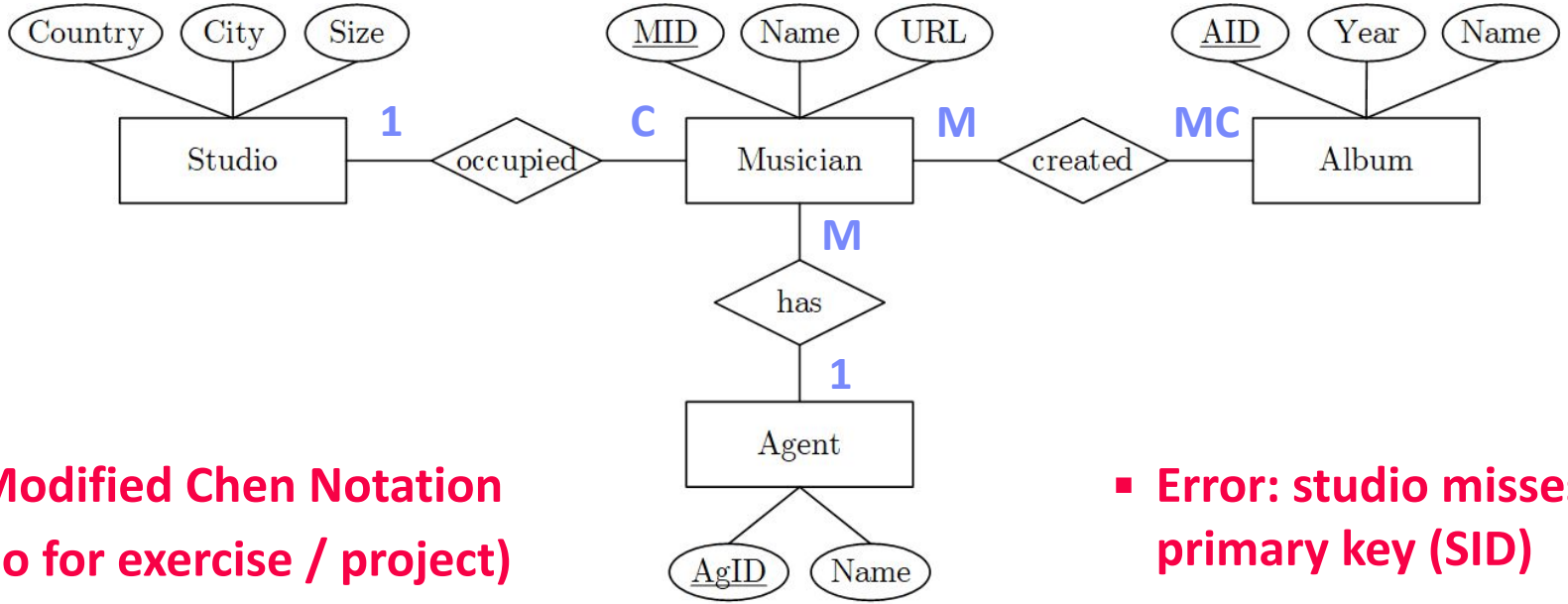


Another Example: Solution



- **Modified Chen Notation (also for exercise)**
- **Do you notice anything else?**

Another Example: Solution

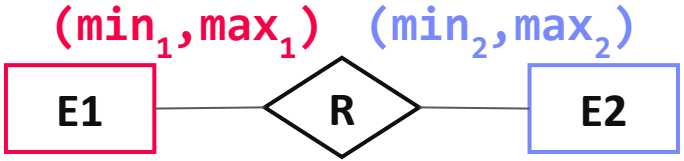


- **Modified Chen Notation (also for exercise / project)**
- **Error: studio misses primary key (SID)**

(min,max)-Notation

Alternative Cardinality Notation

- **Indicate concrete min/max constraints**
  (each entity is part of at least/at most x relationships)
- Modified chen --» **incoming** connection
- (min,max) --» **outgoing** connection
- **Wildcard *** indicates arbitrary many (i.e., MC)



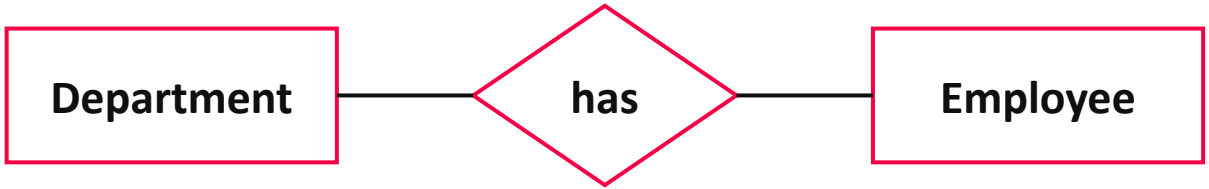
(min,max)-Notation

(each department has 1 – 70 employees)

(1,70)

(each employee in exactly one department)

(1,1)



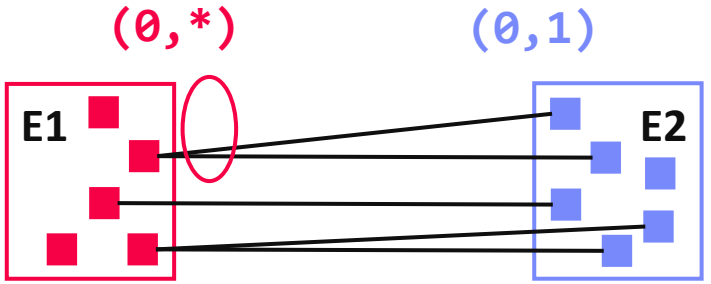
Modified Chen notation

(for comparison)

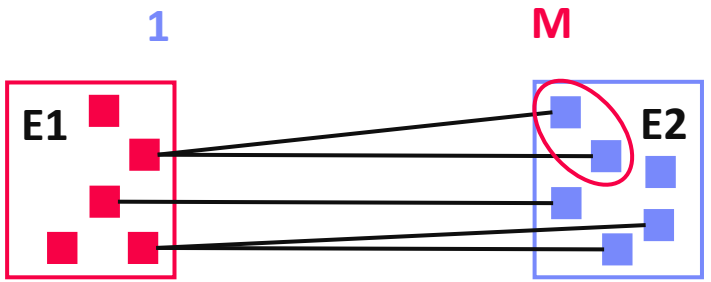
(min,max)-Notation

- **Problem:** Where do these conflicting notations come from?

- Understanding (min, max)-Notation
  - **Focus on relationships!**
  - Describes number of outgoing relationships for each entity



- Understanding Chen- / Modified-Chen-Notation
  - **Focus on entities!**
  - Describes number of target entities (over relationships) for each entity

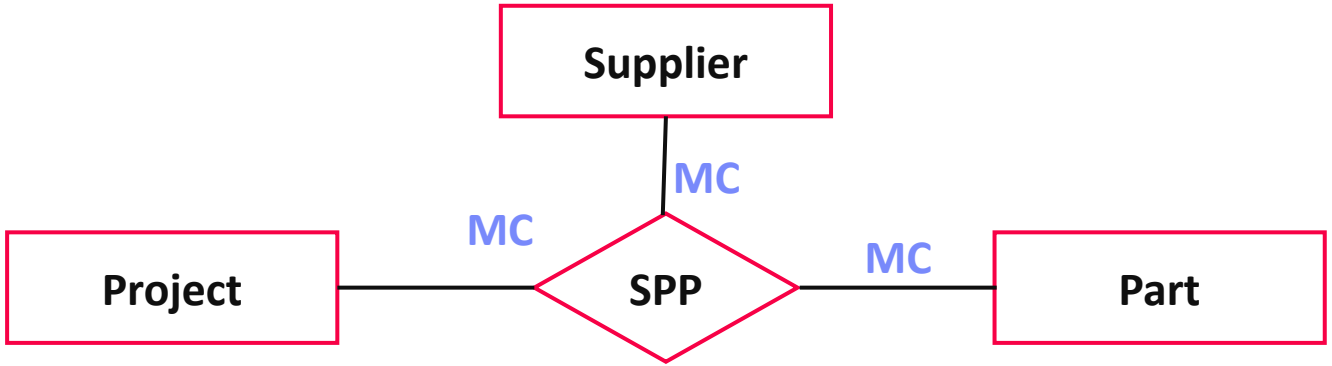


N-ary Relationships

- Relationship among multiple entities
- In some cases N-ary relationship can be **converted to binary relationships**

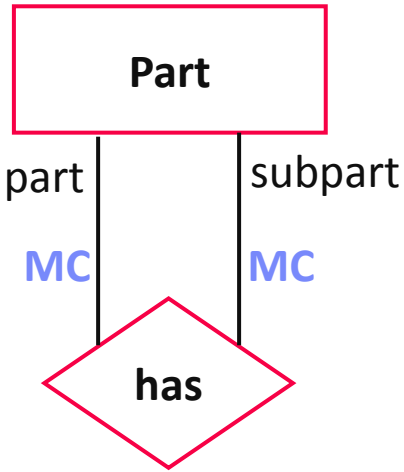
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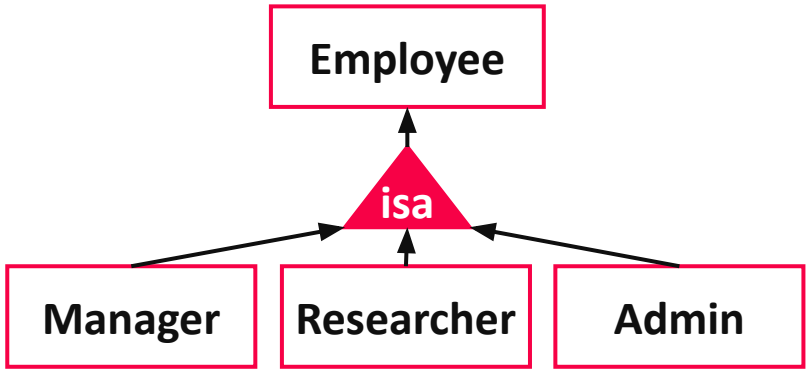
Recursive Relationships

- Definition
  - Recursive relationships are **relations between entities of the same type**
  - Use **roles** to differentiate cardinalities



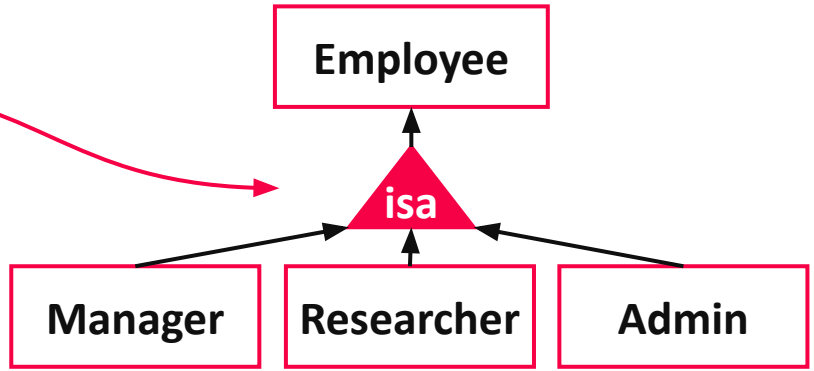
Specialization via Subclasses

- **Tree of specialized entity types**
  Graphical symbol: triangle
- Each **entity of subclass** is **entity of superclass**, but **not vice versa**



Specialization via Subclasses

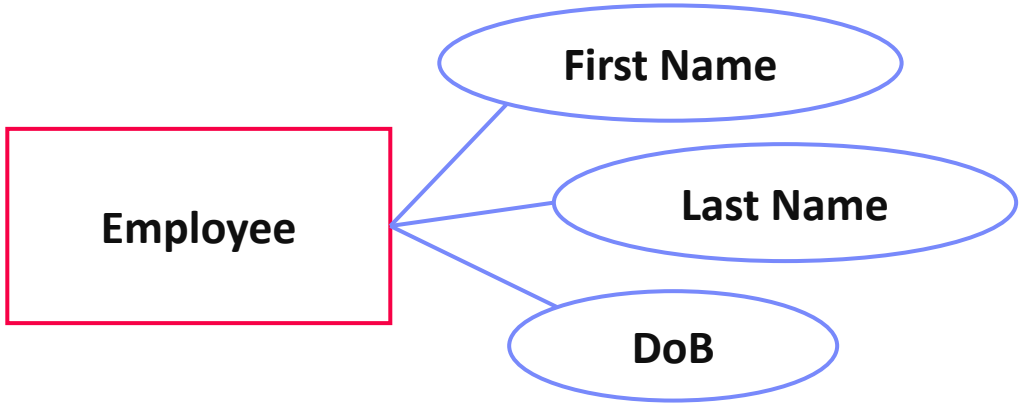
- **Tree of specialized entity types**
  Graphical symbol: **triangle**
- Each **entity of subclass** is **entity of superclass**, but **not vice versa**
- No multi-inheritance



Types of Attributes

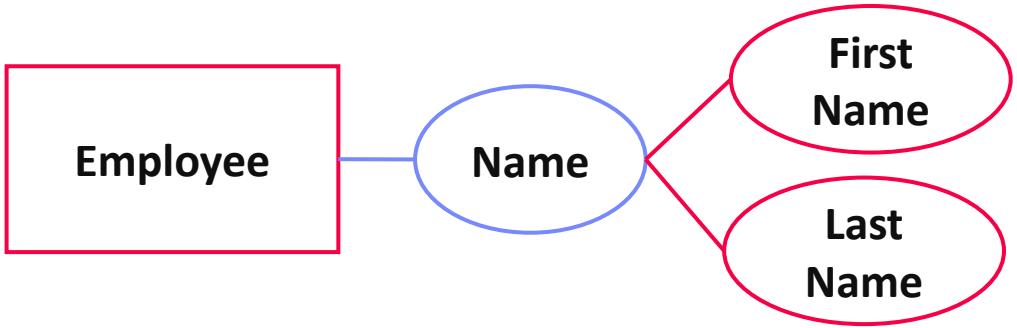
Atomic Attributes

- Basic, **single-valued** attributes



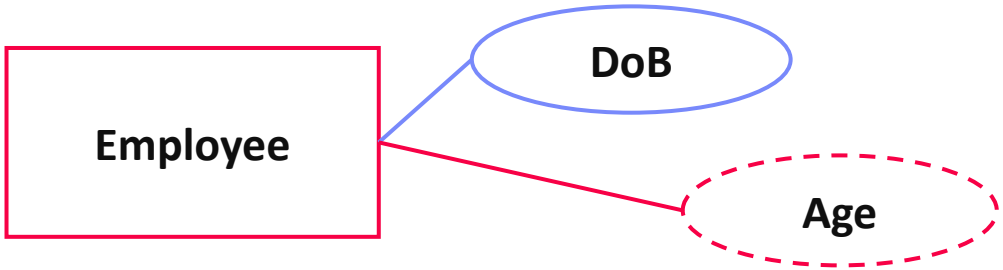
Composite Attributes

- Attributes as **structured data types**
- Can be represented as a **hierarchy**



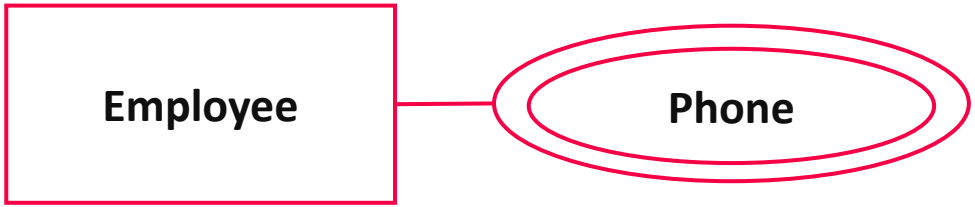
Derived Attributes

- Attributes **derived from other data**
- Examples: Number of employees in dep, employee age, …



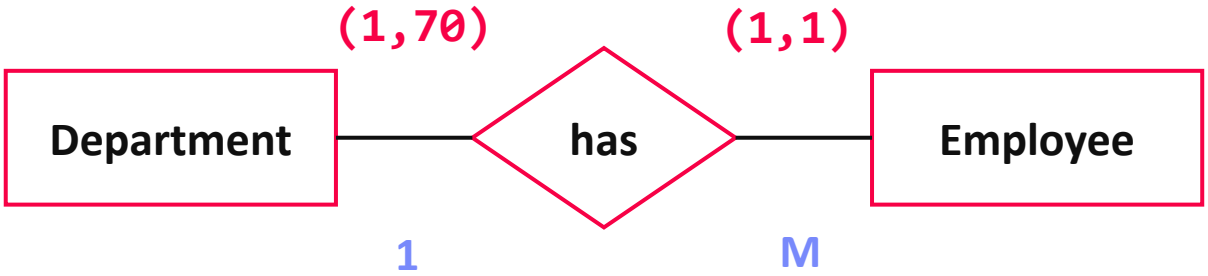
Multi-valued Attributes

- Attributes with list of similar entries



Design Decisions - Notations

- Which notations to use (Chen, Modified Chen, (min,max)-notation)?
  - **Modified Chen for exercise and project!**



Design Decisions - Entities

- What are the **entity types**?
- What are the **attributes of each entity type**?
- What are **key attributes** (one or many)?
- What are **weak entities** (with partial keys)?



Design Decisions - Relationships

- What are the **relationship types between entities** (binary, n-ary)?
- What are the **attributes of each relationship type**?
- What are the **multiplicities /cardinalities**?

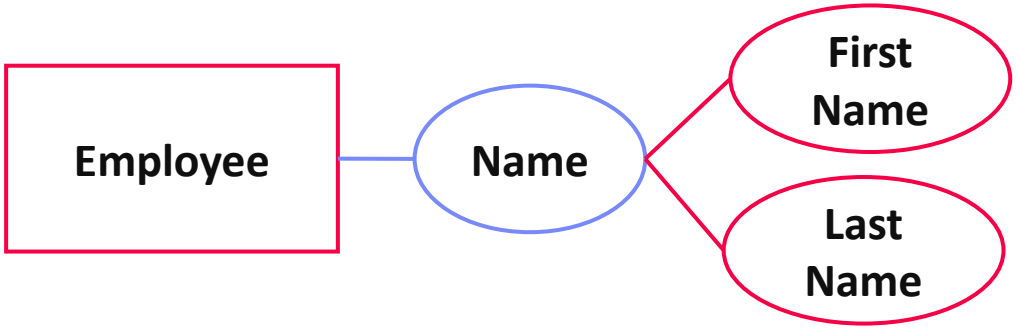
Design Decisions - Relationships

- What are the **relationship types** between entities (binary, n-ary)?
- What are the **attributes of each relationship** type?
- What are the **multiplicities /cardinalities**?

- **Stick to binary** relationships whenever possible
  - Also for exercise and project
  - BUT **sometimes** n-ary relationship make sense

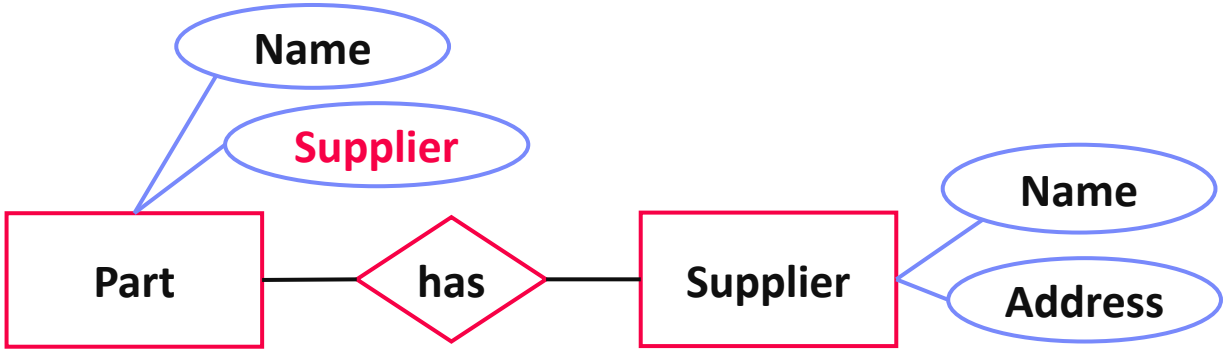
Design Decisions - Attributes

- What are atomic, composite, multi-valued, or derived attributes?
  - **Use atomic** whenever possible

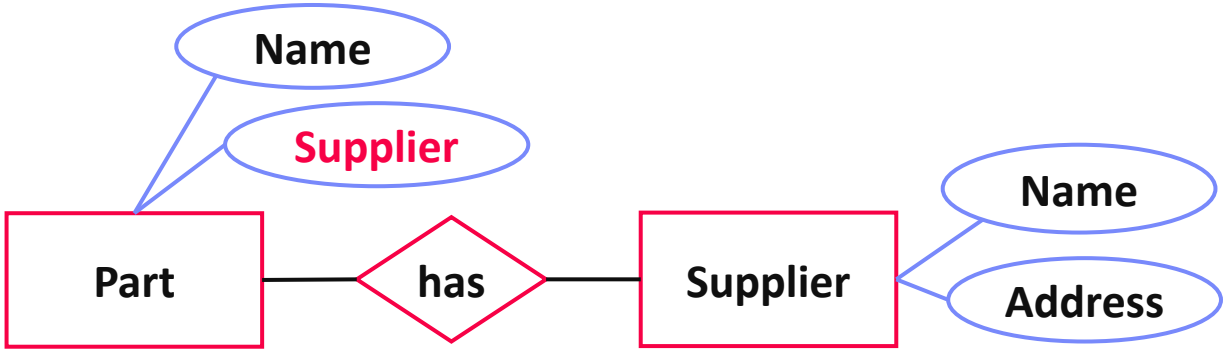


Design Decisions
Examples of Poor Choices

Example 1

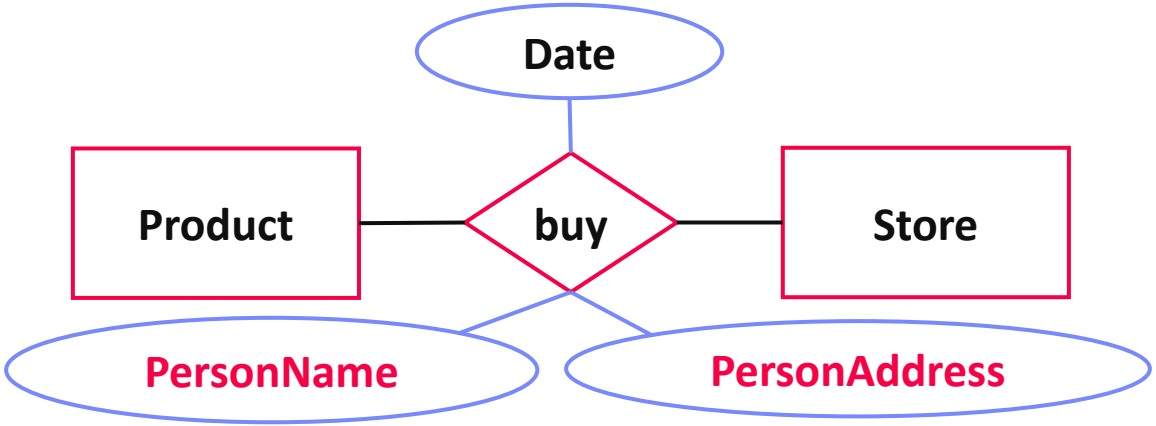


Redundant attributes

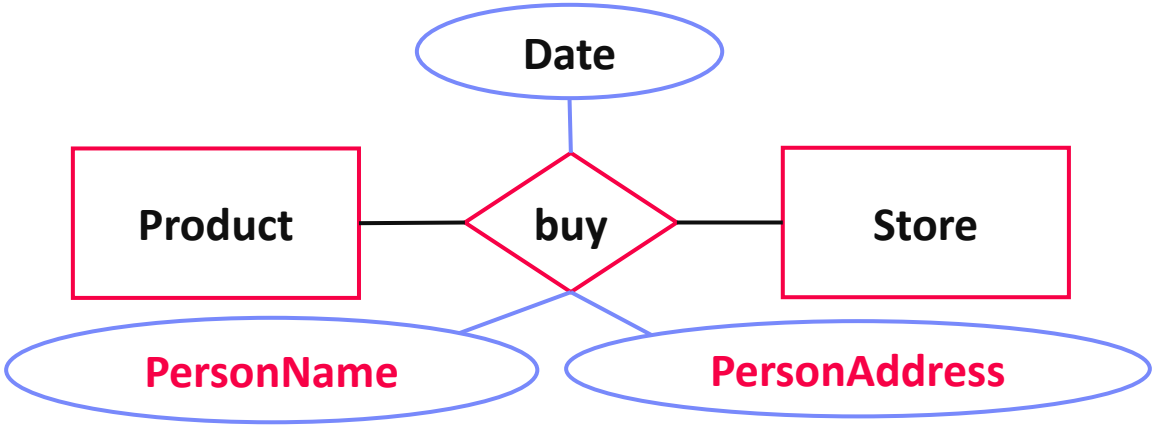


- **Redundant supplier name**
  in Part and Supplier

Example 2



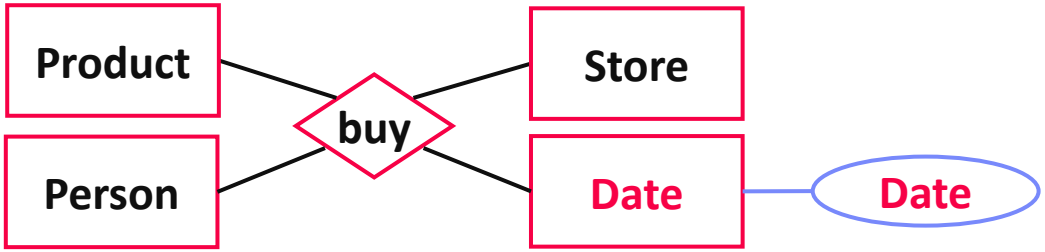
Redundant information



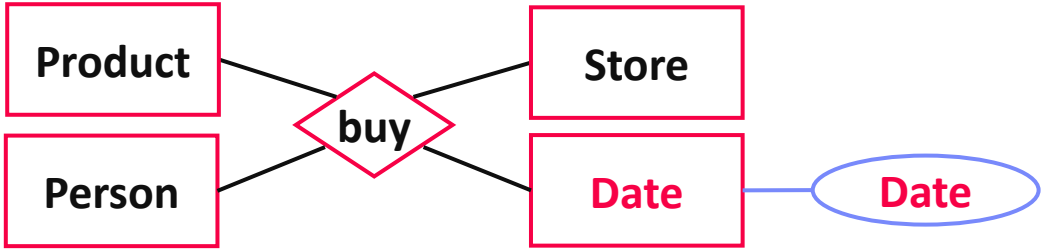
- **Missing person entity type**

→ redundancy per purchase

Example 3

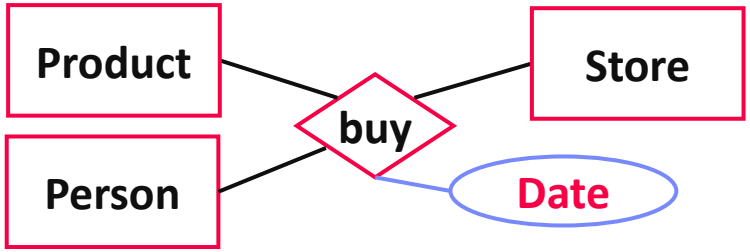


Unnecessary Complexity



- **Unnecessary entity type Date**

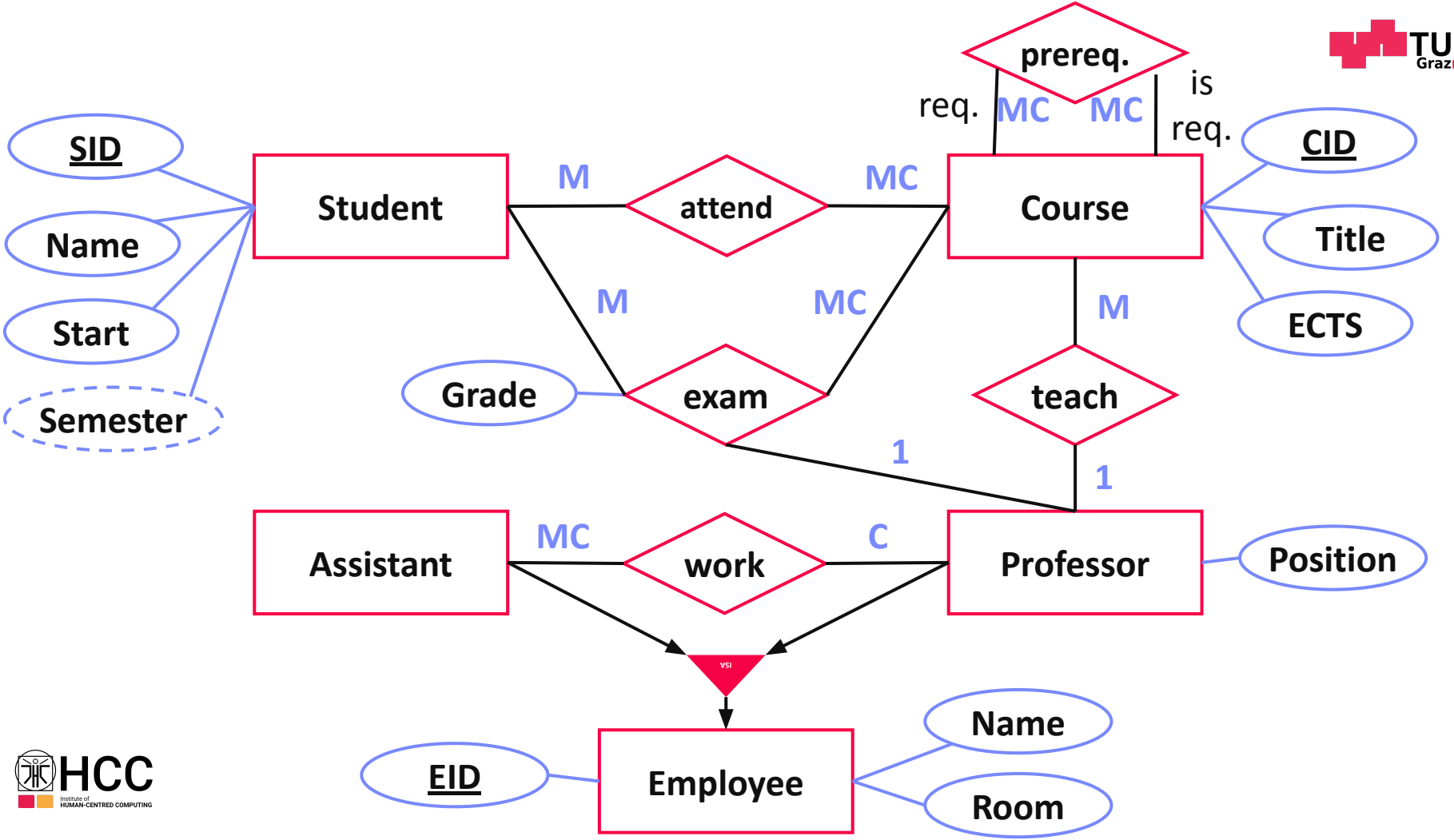
→ Better to store it with the relationship „buy“



A UniversityDB Example

- Discourse of Real Mini World
  - **Students** (with SID, name, start date, and semester) attend **courses** (CID, title, ECTS), and take graded exams per course
  - A **course** may have other courses as prerequisites
  - **Professors** teach and grade courses and have positions (e.g. Dean of Studies), **assistants** may work for professors
  - Both professors and assistants are university **employees** (EID, name, and room number)**; professors** also have a position (e.g. Dean of Studies)

- Task: Create an ER diagram in Modified Chen notation
  - Include entity types, relationship types, attributes, and generalizations
  - Mark primary keys, roles for recursive relationships, and derived attributes


prereq.
req.
MC
MC
is
req.
SID
Student
M
attend
MC
Course
CID
Name
Title
Start
M
MC
M
ECTS
Grade
exam
teach
Semester
1
1
Assistant
MC
work
C
Professor
Position
ISA
EID
Employee
Name
Room

Exercise 1 – Data Modeling

Deadline Tue 31st of March 2026 (23:59)

Exercise 1 Data Modeling (25 points)

- Dataset: **Multitable Ecommerce European Fashion**
  - Dataset from Kaggle
  - Simulates the operation of a fictitious fashion store operating in Europe via E-commerce and Mobile App.
  - Artificial and created for educational purposes.
  - https://www.kaggle.com/datasets/joycemara/european-fashion-store-multitable-dataset
  - **A clean dataset will be provided for the exercises! Don't download the raw one!**

- **Task 1: ER Modeling (15 points)**
- **Task 2: Mapping ER Model to Relational Model (10 points)**



Exercise 1

Task 1.1 ER Modeling (15/25 points)

Create an **ER diagram in Modified Chen (MC) notation** – including entity types, relationship types, attributes, cardinalities, and keys (create surrogate keys if natural unique identifiers are missing) – for managing data stored on a platform selling gift cards of various shops.

Please, use the digital tool **draw.io** for designing the ER diagram. We recommend draw.io since it has a toolbar for ER elements (among others).

Exercise 1

- **Details**
  - The store manages information about products. Each product is described by a unique product id. Additionally, the database stores the product name, brand, category, color, size, gender, cost price, and catalog price (list price). A product may exist in the database without appearing in any sale.

  - The store manages inventory through stock. To uniquely identify a stock entry, the system requires both the product and its storage country. This ensures that inventory quantities are managed separately for each location. The database stores the stock quantity available for each product in each specific country. A single product may have multiple stock entries across different countries, while each stock entry corresponds to exactly one product.

Exercise 1

- **Details (cont.)**
  - The store records sales (orders). Each sale is uniquely identified by a sale id. For each sale, the database records the sale date, the country, and a discounted indicator (specifying if a discount was applied). The total amount of the transaction is calculated from the associated sale items, shipping costs, and discounts. Each sale is associated with exactly one customer, while a customer may be associated with multiple sales. It is important to note that a customer may exist in the database without having any sales recorded.
  - Each sale consists of one or more sale items. Each sale item is identified by a unique
  - item id. This entity links a sale to a product and stores detailed transaction data, including the quantity, the original price, the unit price (price actually paid), the discount applied (absolute value), the discount percent, a discounted indicator, and the final item total. Each sale item refers to exactly one sale and exactly one product.

Exercise 1

- **Details (cont.)**
  - The database contains data on customers registered in the store. Each customer is identified by a unique customer id. Additionally, the database stores the customer's country, age range, and signup date.
  - Sales can be influenced by promotional campaigns. Each campaign is uniquely identified by a campaign id. The database stores the campaign name, the channel (e.g., Email, Social Media), start date, end date, discount type (percentage or fixed amount), and discount value. A sale can be associated with at most one campaign, while a campaign may be associated with many sales. A sale may reference a campaign only if the sale date lies within the campaign period. Furthermore, campaigns may target specific products. A single campaign can promote multiple products, and a product can be associated with multiple campaigns.

Exercise 1

Task 1.2 Mapping ER Diagrams into the Relational Mode (10/25 points)

- Create a relational schema for the ER diagram designed in Task 1.1 and bring it into third normal form. This schema should include the relations and typed attributes, as well as all primary and foreign keys.
- Then, use the PostgreSQL Data Base Management System to create a SQL script (CreateSchema.sql) with CREATE TABLE statements.

Exercise 1

- All details in Exercise Description
- Use only Modified Chen Notation: **M, MC, C, 1.**
- Deliverable 1: **ERDiagram.pdf (1 page PDF)**
- Deliverable 2: **CreateSchema.sql (SQL script)**

Example

Disclaimer: This section contains an example. The exercise to be solved to approve the course can be found in the Teach Center.

Task 1.1

Task 1.2

PostgreSQL & pgAdmin

Example T1.1 (similar to our exercise)

Create an ER diagram in Modified Chen notation for managing data stored on the Formula 1 Championship Management System

The database should capture the essential information of the following statements:

- The database contains data on **drivers** participating in the F1 Championship. Each driver has a unique ID (e.g., 44 for Lewis Hamilton), name, nationality, and date of birth.
- **Teams** participate in the championship, and each team has exactly two drivers. A team is described by its name, country, and foundation year.
- Each **Grand Prix** is an event held in a specific country and has attributes such as circuit name, date, and number of laps. Each Grand Prix can have multiple drivers competing, and each driver may participate in multiple Grand Prix events.






Example T1.1 (similar to our exercise)

Create an ER diagram in Modified Chen notation for managing data stored on the Formula 1 Championship Management System

Step 1: Identify entities and attributes

Entity 1: Driver

- ID (Primary Key)
- Name
- Nationality
- Date of Birth

Entity 2: Teams

- Team Name (Primary Key)
- Country
- Foundation Year

Entity 3: Grand Prix

- Grand Prix ID (Primary Key)
- Circuit Name
- Country
- Date
- Number of Laps

Example T1.1 (similar to our exercise)

Create an ER diagram in Modified Chen notation for managing data stored on the Formula 1 Championship Management System

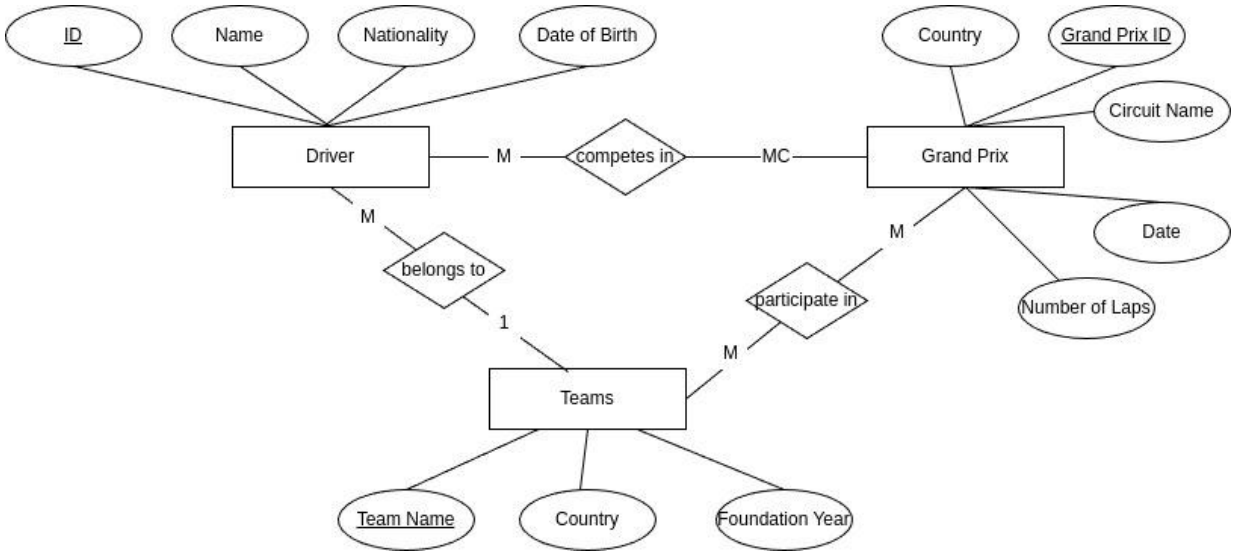
Step 2: Relationships and cardinalities

Relationship	Cardinality	Explanation
Driver - Belongs To - Team	(M:1)	Each driver belongs to one team, and each team has exactly two drivers.
Driver - Competes In - Grand Prix	(M:MC)	A Grand Prix has multiple drivers competing, and a driver may participate in multiple Grand Prix races (or none).
Team - Participates In - Grand Prix	(M:M)	Each team participates in multiple Grand Prix races, and each Grand Prix has multiple teams competing.

Example T1.1 (similar to our exercise)

Create an ER diagram in Modified Chen notation for managing data stored on the Formula 1 Championship Management System

Step 3: ER Map



Example T1.2 (similar to our exercise)

Create a relational schema for the ER diagram designed in Task 1.1 and bring it into third normal form. This schema should include the relations and typed attributes, as well as all primary and foreign keys. Then, use the PostgreSQL Data Base Management System to create a SQL script (CreateSchema.sql) with CREATE TABLE statements.

Step 1:

- Download installer via: https://www.postgresql.org/download/
- Setup PostgreSQL in your Laptop (See Tutorial in Teach Center)

Example T1.2 (similar to our exercise)

Step 2: create your schema and normalize (1NF, 2NF, **3NF**)

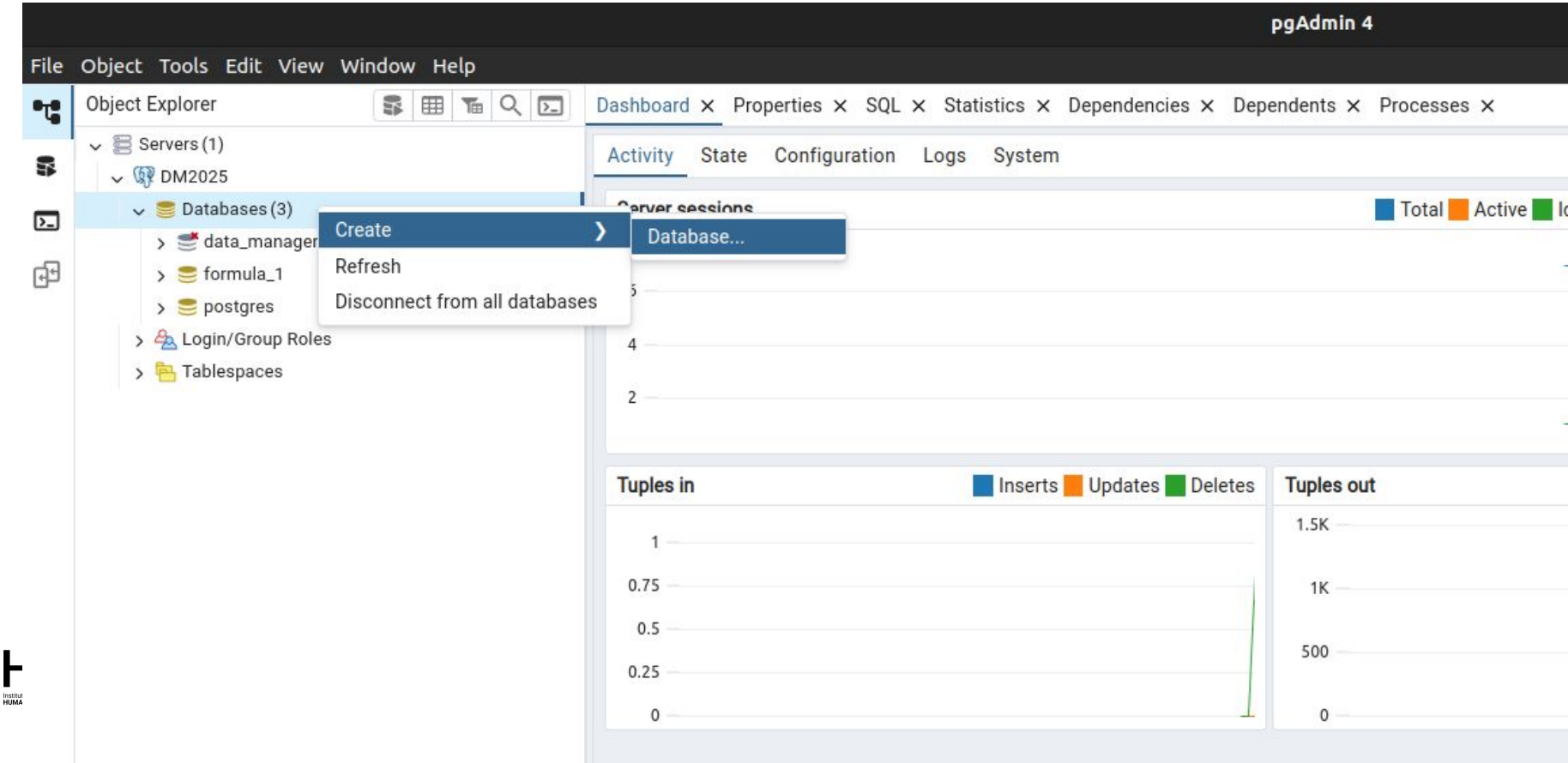
(CREATE Table Statements)

Example T1.2 (similar to our exercise)

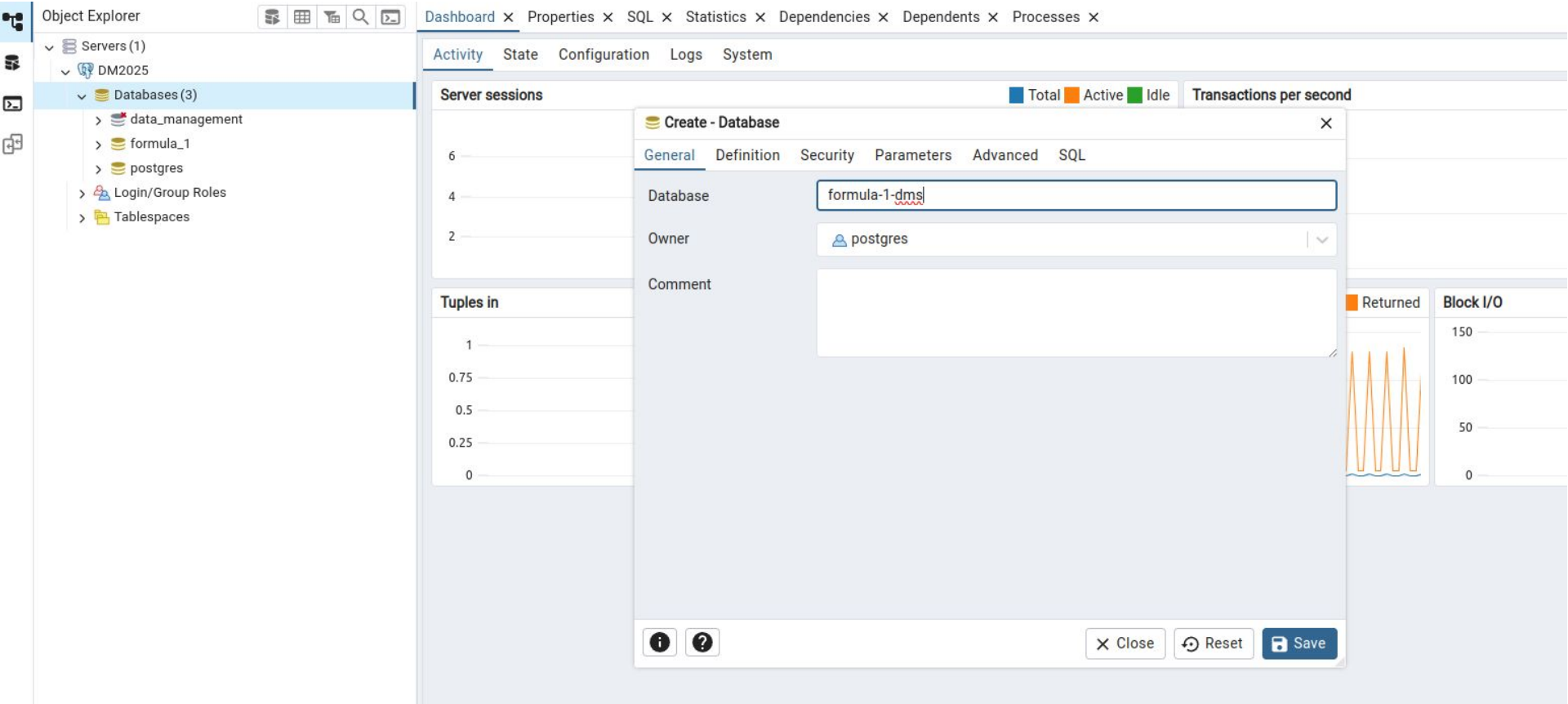
Step 3: Implement your schema in PostgreSQL



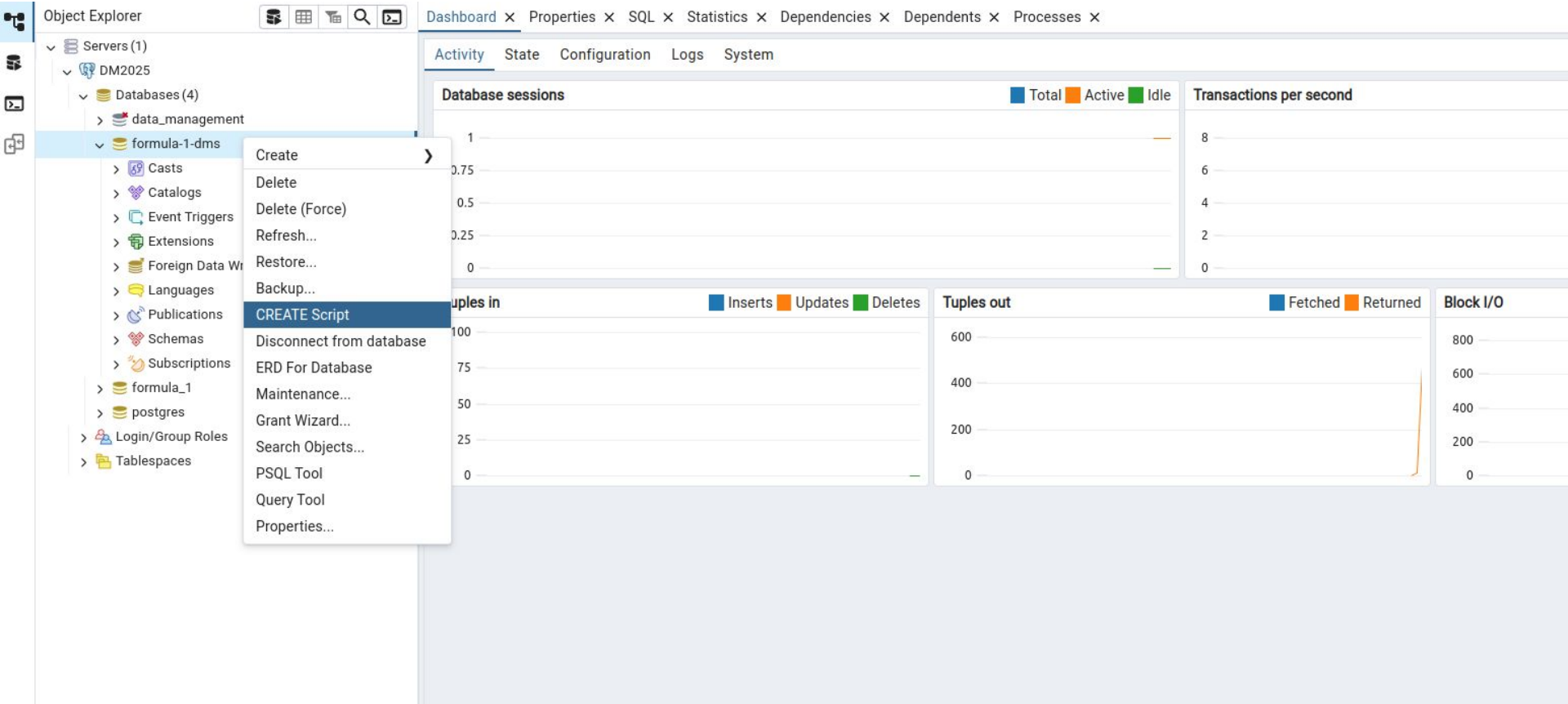
Example T1.2 (similar to our exercise): Create DB



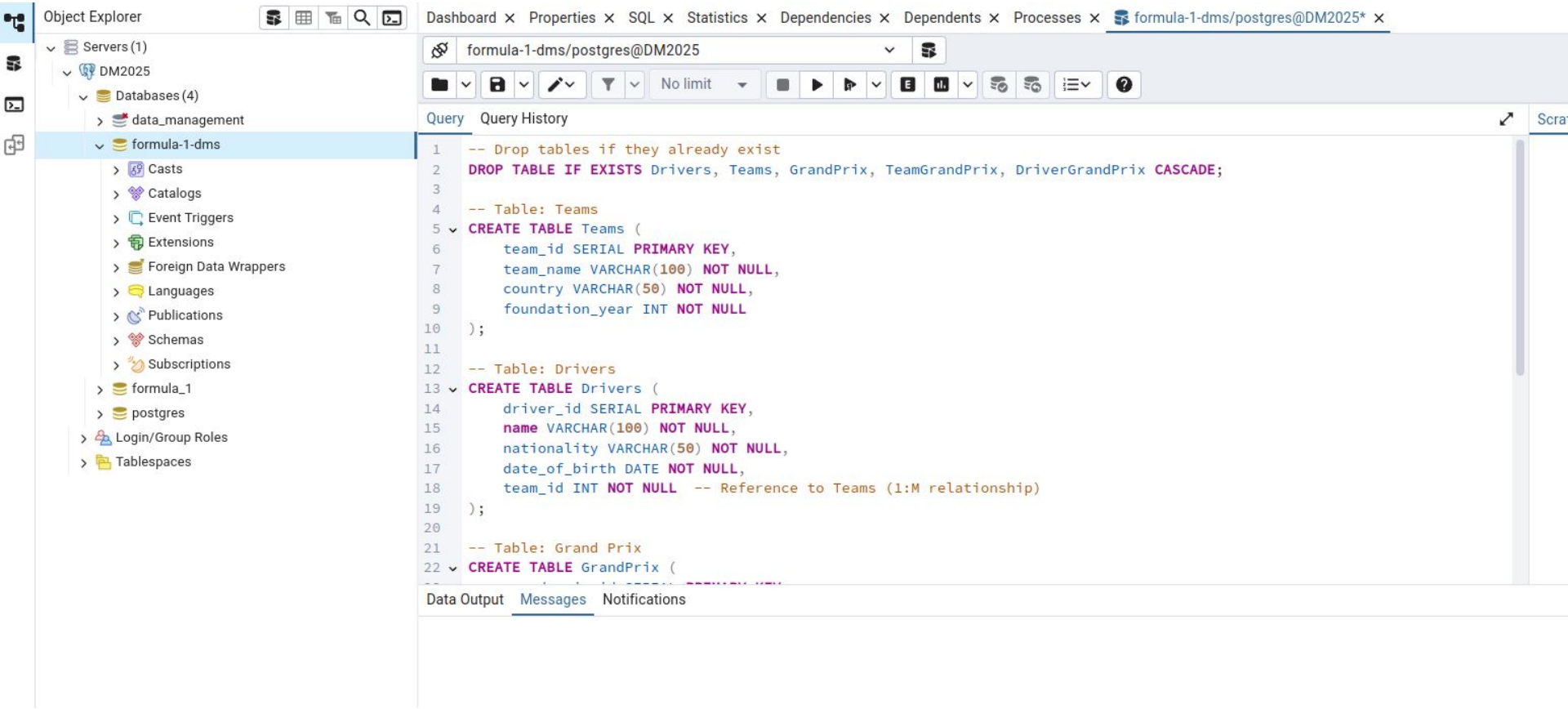
Example T1.2 (similar to our exercise): Create DB



Example T1.2 (similar to our exercise): Create Schema



Example T1.2 (similar to our exercise): Create Schema



Conclusions

- **Summary**
  - **Database design**
  - **Entity-relationship (ER) models and diagrams**
  - **Exercise 1 - T1.1**

Conclusions

- **Summary**
  - Database design
  - Entity-relationship (ER) models and diagrams
  - Exercise 1 - T1.1.

- **Next week**
  - Relational databases and normalization
  - Bit SQL for Exercise 1
  - Exercise 1 - T1.2)